PLC-01
LS XBC-DR32H
MAIN CELL CONTROLLER

16 DIGITAL INPUTS
P00 ~ P0F

16 DIGITAL OUTPUTS
P20 ~ P2F

INPUTS	OUTPUTS
P00 ES-01	P20 MTR-01
P01 PB-01	P21 MTR-01 DIR
P02 PB-02	P22 SOL-01
P03 PB-03	P23 SOL-02
P04 PE-01	P24 SOL-03
P05 PE-02	P25 SOL-04
P06 PE-03	P26 LT-01
P07 DR-01	P27 BZ-01
P08 LS-01	P28~P2F SPARE
P09 LS-02	
P0A LS-03	
P0B~P0F SPARE	

[FIELD DEVICE GROUPING]

[SAFETY / OPERATOR INPUT]
ES-01,
PB-01, PB-02, PB-03,
DR-01

[PROCESS SENSOR INPUT]
PE-01, PE-02, PE-03,
LS-01, LS-02, LS-03

[ACTUATOR OUTPUT]
MTR-01,
SOL-01, SOL-02, SOL-03, SOL-04,
LT-01,
BZ-01

DIGITAL INPUT ALLOCATION

PLC ADDRESS	TAG	DEVICE	NORMAL STATE	PLC FUNCTION
P00	SR-01-AUX	Safety Relay Auxiliary Status	ON = Safety Output Enabled	Safety relay output enabled / safety chain healthy confirmation
P01	PB-01	Start Push Button	OFF	Automatic cycle start request
P02	PB-02	Stop Push Button	OFF	Normal stop request
P03	PB-03	Reset Push Button	OFF	Alarm reset request
P04	PE-01	Infeed Presence Sensor	OFF	Detect package unit at infeed
P05	PE-02	Test Position Sensor	OFF	Confirm package unit at test nest
P06	PE-03	Outfeed Confirm Sensor	OFF	Confirm discharge completion
P07	DR-01-AUX	Guard Door Auxiliary Status	ON = Guard Door Closed	Guard door status monitoring only; not part of safety function
P08	LS-01	Clamp Closed Feedback	OFF	Confirm test clamp closed
P09	LS-02	PASS Gate Position Feedback	OFF	Confirm PASS gate actuated
P0A	LS-03	FAIL Gate Position Feedback	OFF	Confirm FAIL gate actuated
P0B	LS-04	RETEST Gate Position Feedback	OFF	Confirm RETEST gate actuated
P0C	LS-05	Sorter Home Feedback	ON at home	Confirm common sorter mechanism at home position
P0D	SPARE-IN-03	Spare Input	-	Reserved
P0E	SPARE-IN-04	Spare Input	-	Reserved
P0F	SPARE-IN-05	Spare Input	-	Reserved

DIGITAL OUTPUT ALLOCATION

PLC ADDRESS	TAG	DEVICE	PLC OUTPUT FUNCTION	FAIL-SAFE STATE
P20	MTR-01-RUN	Conveyor Motor Run	Conveyor run enable command	OFF
P21	MTR-01-DIR	Conveyor Motor Direction	ON = Reverse / OFF = Forward direction select	OFF; motor disabled by P20
P22	SOL-01	Test Clamp	Clamp close command	OFF / Released
P23	SOL-02	PASS Gate	Route unit to PASS bin	OFF / Home
P24	SOL-03	FAIL Gate	Route unit to FAIL bin	OFF / Home
P25	SOL-04	RETEST Gate	Route unit to RETEST bin	OFF / Home
P26	LT-01-G	Tower Lamp Green	Auto / Run	OFF
P27	LT-01-Y	Tower Lamp Yellow	Manual / Wait / Warning	OFF
P28	LT-01-R	Tower Lamp Red	Alarm / Fault	OFF
P29	BZ-01	Buzzer	Alarm audible indication	OFF
P2A	SPARE-OUT-01	Spare Output	Reserved	OFF
P2B	SPARE-OUT-02	Spare Output	Reserved	OFF
P2C	SPARE-OUT-03	Spare Output	Reserved	OFF
P2D	SPARE-OUT-04	Spare Output	Reserved	OFF
P2E	SPARE-OUT-05	Spare Output	Reserved	OFF
P2F	SPARE-OUT-06	Spare Output	Reserved	OFF

NOTE 1:
MTR-01 shall remain de-energized whenever P20 is OFF, regardless of P21 direction selection state.

NOTE 2:
SOL-02, SOL-03, and SOL-04 shall be mutually exclusive.
Only one bin gate may be actuated during a sorting cycle.

[GENERAL I/O DESIGN NOTES]
[1] Safety-related inputs use normally closed contacts where practical to support open-circuit fault detection.
[2] PLC outputs do not directly supply motor power. Interposing relays, drivers, or contactors shall be used.
[3] All unused I/O points are reserved for future expansion.
[4] The final PLC program shall de-energize all actuator outputs on E-Stop, active alarm, or invalid safety condition.
[5] Field-device wiring and terminal assignment shall be defined in subsequent circuit drawings.

COMMUNICATION INTERFACE SUMMARY

Interface	Connected Device	PLC Role	Physical Specification	Purpose
COM-01	ATE-01 / MCU-01 ESP32-S3	Modbus RTU Slave, ID 1	RS-485, 9600 bps, 8N1	Virtual ATE status/result exchange
COM-02	SCADA-01 LabVIEW PC	XGT Device	RS-232C, 9600 bps, 8N1	SCADA monitoring, logging, command request

NOTE 1:
PLC physical outputs are controlled only after safety and sequence interlock validation.

NOTE 2:
Communication signals are logical data interfaces.
Detailed Modbus Register and XGT Device mapping shall be defined in Drawing 010.

NO.	DATE	REVISION
A	2026-08-10	INITIAL RELEASE

DWG TITLE
EDS Wafer Map Linked Package
Final Test and Multi-Bin Sorting Cell
PLC I/O ALLOCATION & INTERFACE LIST

DRAWN BY: J. H. KO
DATE: 2026-08-10
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SHEET NO: 1 OF 1